R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a) Define axiomatic semantics. [2]
b) Give an attribute grammar for simple assignment statements. [3]
c) What do you mean by precision and range? [2]
d) What is aliasing? What are the problems associated with it? [3]
e) Differentiate between actual and formal parameters. [2]
f) What are the three general characteristics of subprograms? [3]
g) Define abstract data type. [2]
h) What is the purpose of a C++ destructor? [3]
i) How Haskell is different from ML? [2]
j) What is procedural abstraction? Give example. [3]

PART - B

(50 Marks)

2.a) Discuss various programming domains and their associated languages.
b) Describe the basic concept of denotational semantics. [6+4]

OR

3.a) What are the potential benefits of studying programming language concepts?
b) Explain with examples how syntactic design choices affect readability. [5+5]

4.a) What do you mean by binding? Give examples of some of the bindings and their binding times.
b) Evaluate the two approaches for supporting dynamic allocation and deallocation for dynamic length strings. [6+4]

OR

5.a) Explain in detail various design issues of character string types.
b) What are dangling pointers and lost heap-dynamic variables? How are they created? [4+6]

6.a) How co-routines are different from conventional subprograms?
b) Explain type checking technique in parameter passing. [5+5]

OR

7.a) What is a subprogram? Discuss the design issues of subprograms.
b) Write a detailed note on local referencing environments. [5+5]

8.a) What are the various methods of exception handling? Discuss.
b) How message passing is implemented in ADA? Give examples. [5+5]

OR

9.a) Explain how information hiding is provided in an ADA package.
b) Discuss about the basic elements of Prolog with examples. [5+5]

10.a) Explain the important functions of LISP.
b) Discuss the key concepts of scripting languages. [5+5]

OR

11.a) What are the three features of Haskell that makes very different from schema?
b) What are the data types supported in Python? Discuss. [5+5]

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